

# Luciano Correa

Rust Backend Developer · Systems Engineering Student

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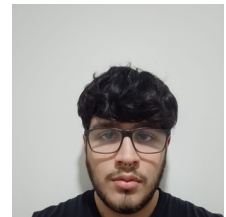
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Buenos Aires, Argentina



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## ABOUT ME

Systems Engineering student at UTN with a strong focus on Rust backend development. Passionate about low-level systems, cryptographic primitives, and peer-to-peer networks. Looking for my first professional opportunity where I can contribute with safe, efficient, and well-designed code.

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## TECHNICAL SKILLS

|                 |                                                               |
|-----------------|---------------------------------------------------------------|
| Languages       | Rust · Go                                                     |
| Async / Runtime | Tokio · async/await · spawn_blocking                          |
| Web / API       | Axum · REST · JSON · HTTP                                     |
| Cryptography    | SHA-256 · Ed25519 · AES-256-GCM · Merkle Tree · Proof of Work |
| P2P Networking  | libp2p · mDNS · Gossipsub · request/response                  |
| Tools           | Git · GitHub · Cargo · serde · thiserror · clap · tracing     |
| Languages       | Spanish (native) · English B2                                 |

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## PROJECTS

### mini-blockchain

*Fully functional blockchain built from scratch in Rust*

- Full blockchain node implementation: cryptography, REST API, and P2P network in a single project.
- Cryptographic core: SHA-256 chain linking, Ed25519 signatures, iterative Merkle tree, and AES-256-GCM wallet encryption.
- REST API with Axum: concurrent read access via `Arc<RwLock>`; mining runs in `spawn_blocking` to never block the async runtime.
- P2P network with libp2p: node discovery via mDNS, chain sync via request/response, block/tx propagation with Gossipsub.
- BIP-39 seed phrase wallet (12 words) with deterministic Ed25519 key derivation.
- Typed error system with `thiserror` and structured logging with tracing.
- Full CLI with `clap`: `new-wallet`, `import-wallet`, `send`, `mine`, `show-chain`, `validate`.
- Stack: Rust 2024 · Tokio · Axum · libp2p 0.54 · serde · sha2 · ed25519-dalek · aes-gcm · clap.

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## EDUCATION

### Bachelor's in Information Systems Engineering

*Universidad Tecnológica Nacional (UTN) · 2nd year, in progress*

Relevant coursework: Algorithms and Data Structures, Computer Architecture, Logic and Discrete Structures.

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## AREAS OF INTEREST

Distributed systems · Applied cryptography · P2P networking · Systems programming · High-performance backend · Open Source